theraCUES™
SIGNALS FOR LIFE

Enabling Translational Research and Precision Medicine

imagine.innovate.impact

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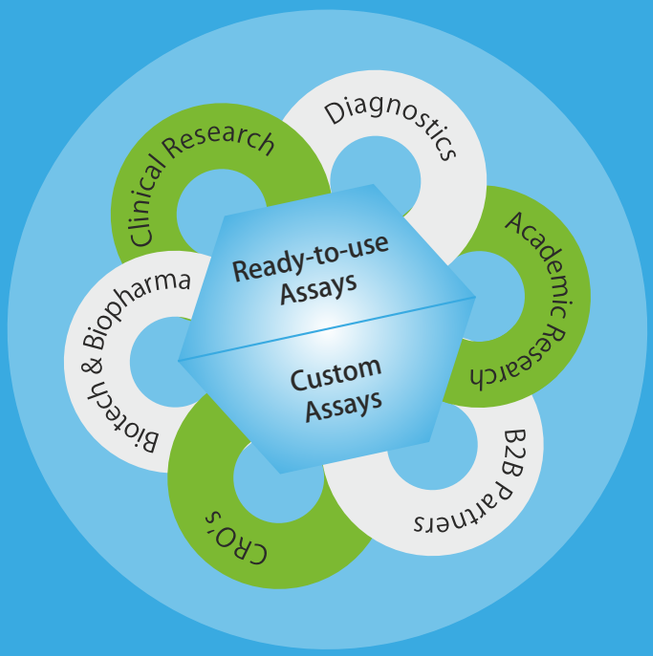
what we do?

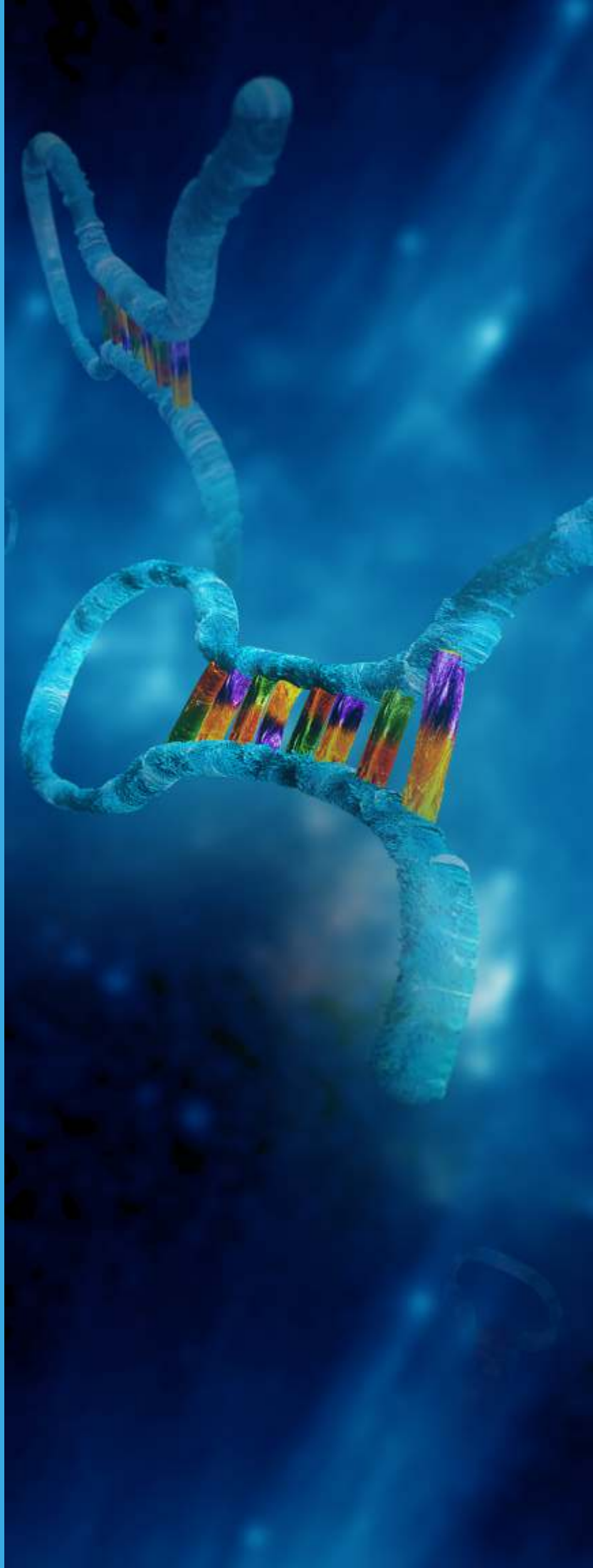
- ✦ Uncover signals (CUES) for targeted therapy (thera)
- ✦ Translate biological research information to actionable assays and screening tests
- ✦ Identify unmet needs and develop amenable solutions for precision medicine



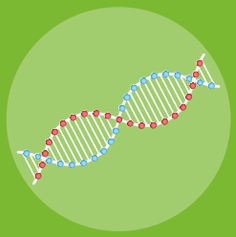
areas we focus...

Diseases where molecular information makes difference in better decision making

Infectious Disease
Neuro Disorders
Oncology
Immuno Oncology
Immune Disorders
Arthritis
Diabetes

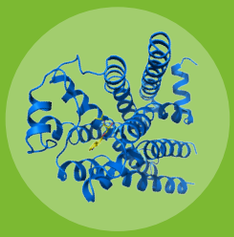


CUES at multiple levels...



Mutations and Structural Variants

Gene Expression and Fusion

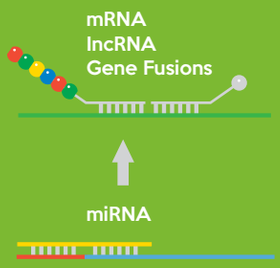


Modulation in Protein Expression

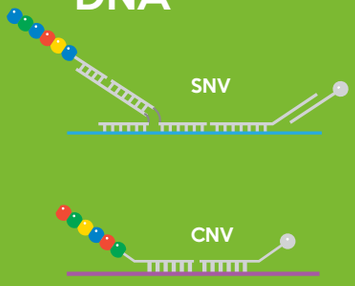
3D BIOLOGY™

leading edge technology, nanoString nCounter, at theraCUES detects signal at all three levels

RNA



DNA



PROTEIN

one chemistry multiple applications

key advantages

- **No PCR**, amplification, or library preparation needed
- **Direct analysis** of tissue, cell and blood lysates, single cells
- **Sample flexibility** - works with low input, fragmented, FFPE or contaminated samples
- **Multiplexing of 10-800 targets** in a single reaction
- Intuitive data analysis with quick turnaround time
- **Pre-designed** or **Customizable panel**
- Proven with 2750+ publications and counting

biomarker CUES - from translational research to precision medicine

profile biological samples and discover biomarkers for oncology and immuno-oncology through our focussed gene expression panels

PanCancer Pathways Panel

770 Cancer Pathways Genes

- 606 pathway genes for 13 canonical pathways
- 124 cancer driver genes
- 40 reference genes

PanCancer Immune Profiling Panel

770 Cancer Immune Genes

- 24 different immune cell types
- 30 CT antigens
- >500 genes covering innate & adaptive immune response
- 40 reference genes

PanCancer Progression Panel

770 Cancer Progression Genes

- 277 angiogenesis genes
- 269 epithelial-mesenchymal transition genes
- 254 extracellular matrix genes
- 173 metastasis genes
- 30 reference genes

discover post-transcriptional regulation using miRNA expression panel

- 800 human miRNAs
- Specific and sensitive profiling
- Direct digital detection without amplification
- Works with FFPE samples

confirm biomarker signatures with our unique protein expression panels

Immune Cell Profiling Panel

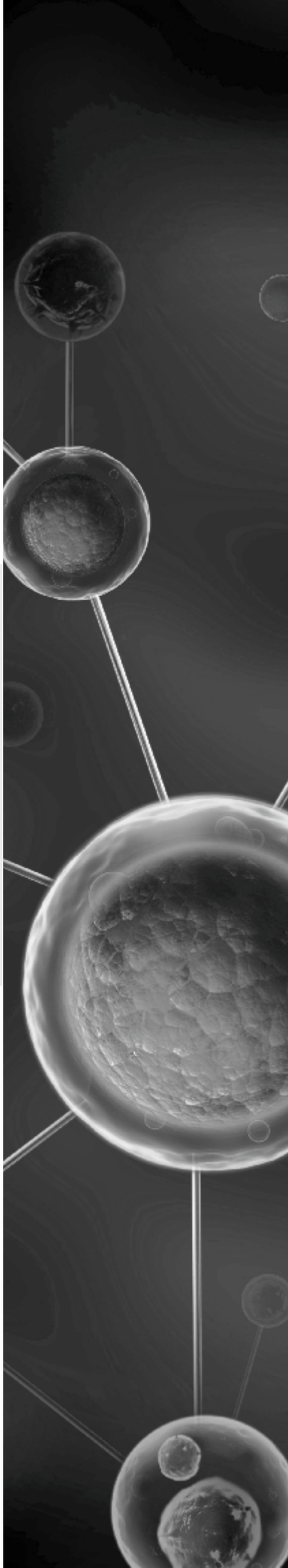
- 30 cell surface markers for comprehensive coverage of immunity cycle
- Works with fresh and cryo-preserved cells

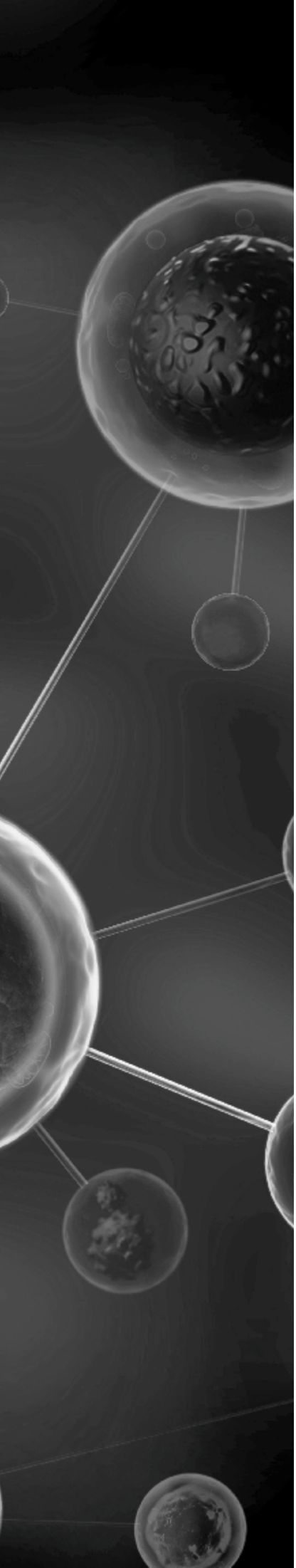
Immune Cell Signalling Panel

- 26 markers comprising of cytokines, chemokines, transcription factors
- Characterization of inflammation and immune effector functions

Solid Tumor Panel

- 28 protein expression to profile key pathways in solid tumor
- Works with 250 ng protein from cell or tissue lysates





uncover somatic mutations and other DNA variants using DNA SNV and CNV panels for fresh / frozen / FFPE samples

95+% sensitivity and specificity for samples with >5% allele frequency

SNV solid tumor assay

104 driver mutations in 25 key solid tumor genes

DNA CNV assay

copy number quantification in 87 genes commonly amplified or deleted in cancer

SNV heme panel

104 driver mutations in 25 key solid tumor genes

detect multiple gene fusions using junction-specific probes

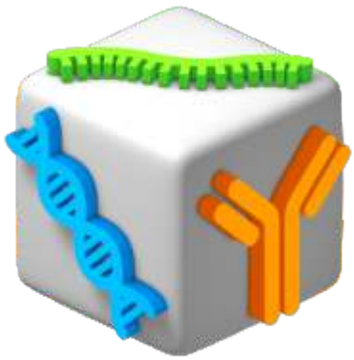
Leukemia

- 27 variants of 9 most common gene fusions in AML, CML, ALL and CLL
- Expression of 12 leukemia genes

Lung Cancer

- 35 fusions for lung cancer
- 24 gene expression imbalance detection

utilize the power of 3D biology using single assays for DNA, RNA and protein detection



3D BIOLOGY™

3D Immuno-oncology assay

770 gene expression, and 30 protein expression probes to profile immune cell types

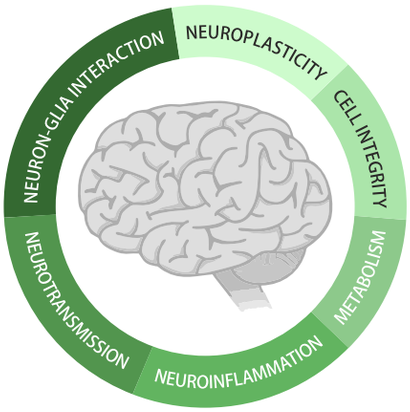
3D Solid tumor assay

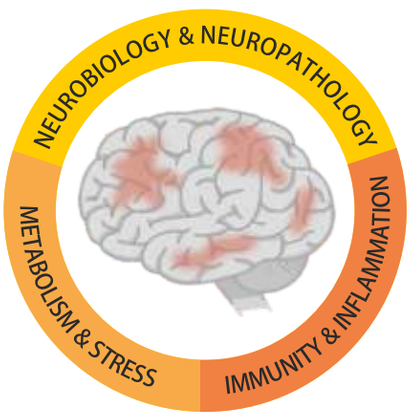
104 actionable SNV and INDELs, 192 gene expression and 28 total and phospho-protein expression probes

3D Heme assay

141 actionable SNV and INDELs, 192 gene expression and 35 total and phospho-protein expression probes

assays in immunology and neurology





Immunology
Expression analysis of 500+ general immunology genes – cytokines and their receptors, chemokines, enzymes, TNF-family, interferons and receptors and anti-fungal response

Inflammation
Expression analysis of 200 genes covering inflammation pathways – apoptosis, EGF, interleukins, Ras, T-cell receptors

Neuropathology
770 gene expression analysis for six major themes of neurodegeneration – for research of Alzheimer's, Parkinson's, ALS etc

Neuroinflammation
770 genes covering 23 neuroinflammation pathways and processes for research on neuropathic pain, traumatic brain injury etc

use the power of nanoString platform to customise and design your own multiplexed assays

- ✦ Probes for your gene list
- ✦ Human or mouse genes
- ✦ Any other organism with available sequence
- ✦ Up to 800 probes in a single assay

Customise Content

Customise Format

- ☑ DNA Variants
- ☑ Gene Expression (mRNA)
- ☑ miRNA Expression
- ☑ Protein Expression
- ☑ All of the above

Attractive Pricing
for custom assays with 24 samples or more

Faster than qPCR. Simpler than NGS.

ready to use nanoString panels available for your translational research

RNA Expression Panels

Cancer
- PanCancer Pathway (H 770 / M 770 genes)
- PanCancer Progression (H 770 / M 770 genes)
- PanCancer Immune profiling (H 770 genes)
- PanCancer IO 360 + Standard Oligos (H 770 / M 770 genes)
- Breast Cancer 360 + Standard Oligos (H 776 genes)
- Cancer metabolism (H 192 genes)
- Cancer reference (H 236 genes)
- Heme – Hematalogic Oncology (H 192 genes)
- Fibrosis (H 770 / M 770 genes)

Pathway Biology
- DNA Repair (H 192 genes)
- MAPK-PI3K Pathways (H 192 genes)
- WNT Pathways (H 192 genes)
- Kinase Pathway (H 536 genes)
- Notch Signalling Pathway (H 24 genes)

Neurology
- Neuropathology (H 770 / M 770 genes)
- Neuroinflammation (H 770 / M 770 genes)
- Alzheimer's Disease (H 770 / M 770 genes)

Immune Response
- Immunology (H 594 / M 561 genes)
- Inflammation (H 255 / M 254 genes)
- Myeloid innate immunity (H 770 / M 754 genes)
- Adaptive immunity (H 192 genes)
- Innate immunity (H 192 genes)
- Autoimmune (H 770 / M 770 genes)
- Non Human Primate immune panel (770 genes)
- Organ Transplant Panel (H 770 genes)

Cell Characterization
- CAR-T Characterization (H 770 genes)

Signaling Response
- Intracellular signalling (H 192 genes)
- Cellular profiling (H 192 genes)

Cellular Metabolism
- Metabolic Pathways (H 768 genes)

Protein Panels

- Immune Cell Profiling for cell suspension (H 30 proteins)
- Immune Cell Signaling for cell suspension (H 30 proteins)
- Solid Tumor Signaling for cell lysates (H 28 proteins)
- Solid Tumor Signaling for FFPE (H 26 proteins)
- Heme for Lysate (H 34 proteins)
- Heme for FFPE (H 35 proteins)

DNA Panels

- SNV (Human Solid Tumor, 104 actionable variations)
- Heme (Human 141 actionable mutations)
- CNV (Human 87 cancer specific)
- Karyotype (Human 24 chromosomes abnormalities)

Fusion Panels

- Leukemia Gene Fusion (Human 42 probes, 27 fusion, 12 leukemia biomarkers)
- Lung Gene Fusion (Human 63 probes; 35 fusion, 24 gene expression imbalance)

miRNA Panels

- v3 Human miRNA Panel (800 miRNAs)
- v1.5 Mouse miRNA Panel (>600 miRNAs)
- v1.5 Rat miRNA Panel (>400 miRNAs)

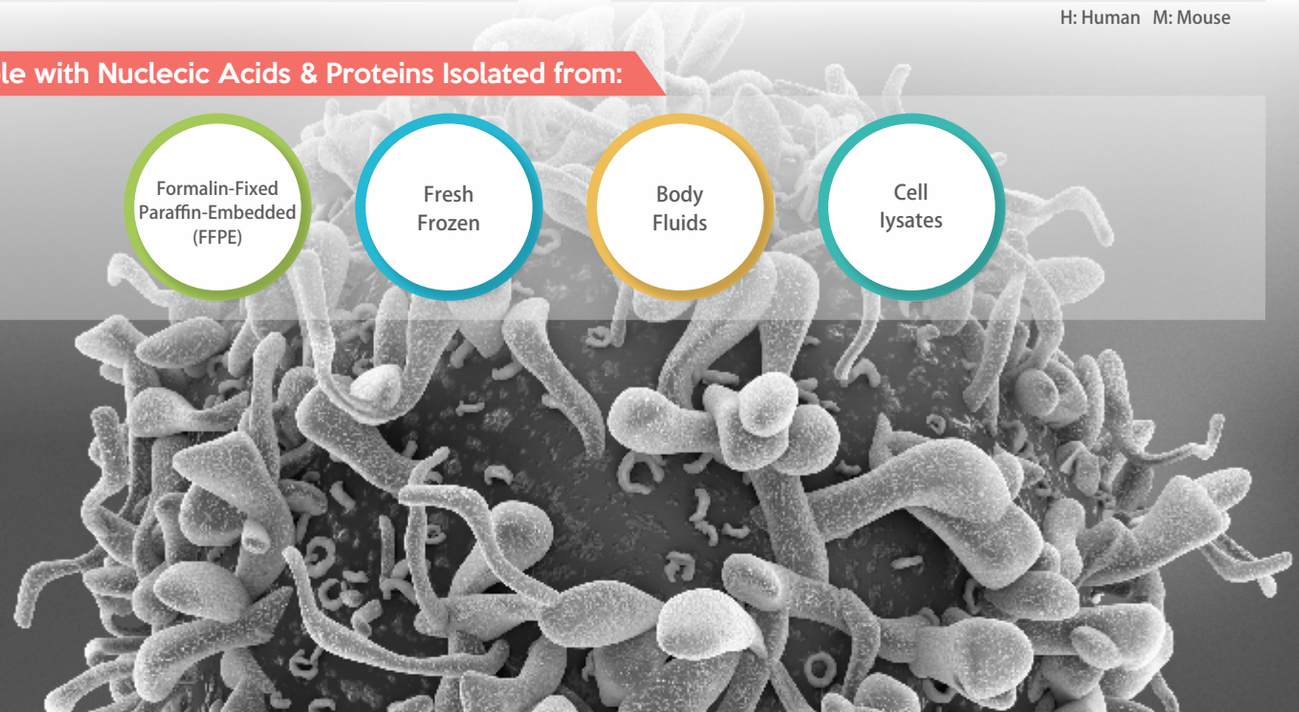
Customer Published Panels

- Subtypes of Diffuse Large B-cell Lymphoma
- Prognosis in Hodgkin's Lymphoma
- Subtypes of Medulloblastoma
- Transcriptional Signatures in Prostate Cancer
- miRNA for Endometriosis and Ovarian Cancer
- Sarcoma Fusion detection panel
- Autoimmune Consortium Discover panel (770 genes)

H: Human M: Mouse

Compatible with Nucleic Acids & Proteins Isolated from:

- Formalin-Fixed Paraffin-Embedded (FFPE)
- Fresh Frozen
- Body Fluids
- Cell lysates



ready to use nanoString panels available for your translational research

Multianalyte panels (RNA:Protein and DNA:RNA:Protein)

2D Immuno-Oncology assays

- RNA: Protein Immune Cell Profiling Assay for cell suspensions:
  PanCancer Immune Profiling (H 770 genes) & Extracellular immunology protein targets (H 30 proteins)

- RNA: Protein Immune Cell Signaling Assay for cell suspensions:
  PanCancer Immune Profiling (H 770 genes) & Intracellular immunology protein targets (H 26 proteins)

2D/3D Solid Tumor Assays

- RNA:Protein Solid Tumor for Lysates:
  RNA expression of PanCancer Pathways (H 770 genes) and solid tumor signaling protein targets (H 28 proteins).

- RNA:Protein Solid Tumor for FFPE:
  RNA expression of PanCancer Pathways (H 770 genes) and solid tumor signaling protein targets (H 26 proteins).

- DNA:RNA:Protein Solid Tumor for Lysates: Solid tumor actionable mutations (104 mutations over 25 genes),
  RNA expression of MAPK-PI3K Pathways (H 192 genes) and solid tumor signaling protein targets (H 28 proteins).

- DNA:RNA:Protein Solid Tumor for FFPE: Solid tumor actionable mutations (104 mutations over 25 genes),
  RNA expression of MAPK-PI3K Pathways (H 192 genes) and solid tumor signaling protein targets (H 26 proteins).

3D Lung Tumor Assays

- DNA:Fusion:Protein Lung Assay for Lysates: Solid tumor actionable mutations (104 mutations over 25 genes),
  Fusion detection (63 probes; fusion, imbalance and reference) and solid tumor signaling protein targets (H 28 proteins).

- DNA:Fusion:Protein Lung Assay for FFPE : Solid tumor actionable mutations (104 mutations over 25 genes),
  Fusion detection (63 probes; fusion, imbalance and reference) and solid tumor signaling protein targets (H 26 proteins)

3D Heme Assays

- DNA:RNA:Protein Heme Assay for Lysate: Liquid tumor/haematologic oncology relevant 141 SNV and INDELS covering 45 genes,
  192 RNA including 12 reference genes, and 34 proteins

- DNA:RNA:Protein Heme Assay for FFPE: Liquid tumor/haematologic oncology relevant 141 SNV and INDELS covering 45 genes,
  192 RNA including 12 reference genes, and 34 proteins

H: Human M: Mouse

theraCUES is the First and Only partner for nanoString services in India

For services and solutions please write to bd@theracues.com or cues@theracues.com

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SIGNALS FOR LIFE

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